Practice Questions

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Score: 38% Passing Score: 80%

Question 1: ✔ Correct

Your client has acquired several small companies and would like to connect them together into one network. Not all of the routers are Cisco devices, and compatibility is a concern.

Which WAN encapsulation method should you recommend your client use?

- Ethernet
- Cisco HDLC
- PAP
- ➡ PPP

EXPLANATION

The point-to-point (PPP) protocol is not proprietary. For this reason, it is the best choice for connecting dissimilar vendor devices.

Cisco HDLC is the default serial encapsulation method, but it is only supported on Cisco devices. PAP is a PPP authentication protocol. Ethernet is not a WAN protocol.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 2: ✕ Incorrect

RouterA is connected to RouterB through Serial1. You want to configure the link to use PPP with CHAP authentication with a password of **cisco**. Which set of commands would you use on RouterA to complete the configuration?

```
RouterA(config)#int s1
RouterA(config-if)#encap ppp
RouterA(config-if)#ppp auth chap
RouterA(config-if)#ppp chap username RouterB password
cisco
```

```
RouterA(config)#int s1
RouterA(config-if)#encap ppp
RouterA(config-if)#ppp auth chap password cisco
```

```
RouterA(config)#int s1
RouterA(config-if)#encap ppp
RouterA(config-if)#ppp auth chap
RouterA(config)#username RouterB password cisco
```

```
RouterA(config)#int s1
RouterA(config-if)#encap ppp
RouterA(config-if)#ppp chap username RouterA password
cisco
```

```
RouterA(config)#int s1
RouterA(config-if)#encap ppp auth chap password cisco
```

EXPLANATION

To complete this configuration, you need to:

1. Identify the interface you want to configure with the **int s1** command.
2. Set PPP encapsulation with the **encapsulation** command.
3. Set the PPP authentication to CHAP with the **ppp authentication** command.
4. Identify RouterB and the password with the **username** command.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 3: ✕ Incorrect

Which of the following statements about the functionality of LCP are true? (Select three.)

- [ ] LCP is responsible for logical addressing in PPP.
- [ ] LCP provides remote terminal access in public data networks.
- [x] ~~LCP encapsulates multiple protocols.~~
- [x] LCP provides multilink support.
- [x] Usernames and passwords may be required during the handshake.
- [ ] Data can be compressed at the source and decompressed at the destination.

EXPLANATION

The link control protocol (LCP) is a sublayer within the PPP protocol stack. LCP options provide authentication, compression, error detection, and multilink functionality.

Multiple protocols are encapsulated in PPP by NCP. LCP is not a Layer 3 protocol, so it does not provide logical addressing. Remote terminal access is a function of the X.25 protocol.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 4: ✕ Incorrect

Which of the following are benefits of LCP? (Select three.)

- [x] Monitors data dropped on the link and avoids frame looping
- [ ] Provides load balancing across multiple links
- [x] ~~Provides physical addressing for the data link layer~~
- [ ] Provides logical addressing for the data link layer
- [x] Negotiates the use (or lack) of authentication before starting the session

EXPLANATION

Benefits of LCP include the following:

- Negotiates the use of authentication before starting the session.
- Monitors data dropped on the link and avoids frame looping (error detection).
- Compresses data at the source and decompresses data at the destination (compression).
- Provides load balancing across multiple links (multilink).

LCP does not encapsulate protocols; NCP handles that task.

LCP is a data link layer protocol, so it does not provide support for physical interfaces. Because PPP is a point-to-point protocol, no logical or physical addressing is necessary.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 5: ✕ Incorrect

What connection order would two TCP/IP routers use to open a session with PPP?

- ~~NCP, authentication, LCP~~
- LCP, authentication, NCP
- NCP, LCP, authentication
- LCP, NCP, authentication

EXPLANATION

PPP uses the following process to open a session:

1. Exchange LCPs to establish the link and negotiate communication parameters.
2. Perform authentication (optional).
3. Exchange NCPs to negotiate the Network layer protocols to use.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 6: ✔ Correct

PPP supports authentication, compression, and multiple Network layer protocols.

Which of the following correctly sequences these functions when a PPP link is established?

- ➡ Negotiate compression settings, perform authentication, negotiate Network layer protocols
- Negotiate compression settings, negotiate Network layer protocols, perform authentication
- Perform authentication, negotiate compression settings, negotiate Network layer protocols
- Negotiate Network layer protocols, perform authentication, negotiate compression settings
- Negotiate Network layer protocols, negotiate compression settings, perform authentication
- Perform authentication, negotiate Network layer protocols, negotiate compression settings

EXPLANATION

PPP uses the following process to open a session:

1. Exchange LCPs to establish the link and negotiate communication parameters (such as compression settings).
2. Perform authentication (optional).
3. Exchange NCPs to negotiate the Network layer protocols.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 7: ✔ Correct

Which of the following protocols is used by PPP to enable support for multiple Network layer protocols?

- SAP
- HDLC
- ➡ NCP
- LCP
- TCP

EXPLANATION

PPP uses the network control protocol (NCP) to support multiple upper-layer protocols. LCP is used to establish the link, negotiate compression and authentication, detect errors, and tear down the link.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 8: ✕ Incorrect

Two routers with the host names SLC and PROVO have been configured to connect using PPP with CHAP authentication through their BRI0 interfaces. Attempts to establish a session between the two routers fail. You check the running configuration on both routers and find the output shown below:

```
hostname SLC
enable password cisco
username PROVO password vanilla
!
!
interface Serial0
ip address 172.16.55.129
255.255.255.252
encapsulation ppp
ppp authentication chap
!
! - remaining output omitted --
```

```
hostname PROVO
enable password ccna
username SLC password chocolate
!
!
interface Serial0
ip address 172.16.55.130
255.255.255.252
encapsulation ppp
ppp authentication chap
!
! - remaining output omitted --
```

What should you do to correct the problem?

- On SLC, change the username to **SLC**. On PROVO, change the username to **PROVO**.
- ~~On PROVO, change the enable password to **cisco**.~~
- On PROVO, change the IP address to 172.16.55.128.
- On SLC, change the username password to **ccna**. On PROVO, change the username password to **cisco**.
- ➡ On SLC, change the username password to **chocolate**.

EXPLANATION

The username passwords used by each router must match. In this scenario, changing the username password on SLC to **chocolate** would correct the problem (you could also change the password on PROVO to **vanilla**).

The username configured on each router must match the host name of the remote router that it will be connecting to. The IP addresses assigned to the interfaces are both on the 172.16.55.128 subnet. You cannot assign that address to a host.

REFERENCES

LabSim for Network Pro, Section 11.2.

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